STAT230A Lecture 13 Frisch-Waugh Lovell Theorem (Application 1)

1 FWL Theorem 的 Application: QR Decomposition

1.1 Corollary: X_1 与 X_2 Orthogonal 的情况

若 $X_1^\top X_2 = 0_{k\times l}$, 即 X_1 的每个 columns 均与 X_2 的所有 columns 正交, 则

$$\tilde{X}_2 = (I - H_1)X_2 \qquad (\text{columns of } X_2 \in \text{Col}(X_1)^\perp)$$

此时,

$$\tilde{\beta}_2 = \hat{\beta}_2$$

Remark

若 X 的所有 columns 均 orthogonal to each other, 则 $\hat{\beta}_j$ 等于 simple regression coefficient for X_j

Logic

根据该 corollary, 对于任意 X, 我们可以试图将其 columns 变为 orthogonal, 随后使用 simple regression 即可

事实上, 这也是 `lm()` 所做的 (QR decomposition), 从而避免求解 $X^\top X$ 的 inverse

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Logic

我们接下来考虑如何得到 orthogonal matrix, 即我们希望将 $X \in \mathbb{R}^{n\times p}$ 转化为 $Q \in \mathbb{R}^{n\times p}$, 其满足:

- $Q^\top Q = I_p$
- Q 的 columns 为 X 的 columns 的 linear combination

一种实现该目的的方法是 Gram-Schmidt Process

1.2 Gram-Schmidt Process

Step 1: 构造 orthogonal vectors

利用 linear regression 构造 orthogonal vectors $\{\vec{U}_1, \vec{U}_2, \dots, \vec{U}_p\}$:

$$\begin{aligned}
&\vec{U}_1 = \vec{X}_1 \\
&\vec{U}_2 = \vec{X}_2 - \hat{\beta}_{X_2\sim U_1}\vec{U}_1 && (\text{确保 } \vec{U}_1 \perp \vec{U}_2) \\
&\vec{U}_3 = \vec{X}_3 - \hat{\beta}_{X_3\sim U_1}\vec{U}_1 - \hat{\beta}_{X_3\sim U_2}\vec{U}_2 && (\text{确保 } \vec{U}_3 \perp \vec{U}_1, \vec{U}_2) \\
&\ \vdots \\
&\vec{U}_p = \vec{X}_p - \hat{\beta}_{X_p\sim U_1}\vec{U}_1 - \cdots - \hat{\beta}_{X_p\sim U_{p-1}}\vec{U}_{p-1} && (\text{确保 } \vec{U}_p \perp \vec{U}_1, \dots, \vec{U}_{p-1})
\end{aligned}$$

Step 2: Normalization

将 orthogonal vectors $\{\vec{U}_1, \vec{U}_2, \dots, \vec{U}_p\}$ normalize 为 unit vectors $\{Q_1, \dots, Q_p\}$:

$$Q_j = \frac{\vec{U}_j}{\sqrt{\vec{U}_j^\top \vec{U}_j}} = \frac{\vec{U}_j}{\|\vec{U}_j\|}$$

由此,

$$Q^\top Q = I \qquad \mathrm{Col}(Q) = \mathrm{Col}(X)$$

Summary: QR Decomposition

综合以上步骤, 我们对 X 进行了以下变换:

$$\begin{aligned}
X &= \begin{bmatrix}\vec{X}_1 & \cdots & \vec{X}_p\end{bmatrix} \\
&= \begin{bmatrix}\vec{U}_1 & \cdots & \vec{U}_p\end{bmatrix}\begin{bmatrix}1 & \hat{\beta}_{X_2\sim U_1} & \hat{\beta}_{X_3\sim U_1} & \cdots & \hat{\beta}_{X_p\sim U_1} \\ 0 & 1 & \hat{\beta}_{X_3\sim U_2} & \cdots & \hat{\beta}_{X_p\sim U_2} \\ 0 & 0 & 1 & \cdots & \hat{\beta}_{X_p\sim U_3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1\end{bmatrix} \\
&= \begin{bmatrix}\vec{Q}_1 & \cdots & \vec{Q}_p\end{bmatrix}\begin{bmatrix}\|\vec{U}_1\| & & \\ & \ddots & \\ & & \|\vec{U}_p\|\end{bmatrix}\begin{bmatrix}1 & \hat{\beta}_{X_2\sim U_1} & \hat{\beta}_{X_3\sim U_1} & \cdots & \hat{\beta}_{X_p\sim U_1} \\ 0 & 1 & \hat{\beta}_{X_3\sim U_2} & \cdots & \hat{\beta}_{X_p\sim U_2} \\ 0 & 0 & 1 & \cdots & \hat{\beta}_{X_p\sim U_3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1\end{bmatrix} \\
&=: QR
\end{aligned}$$

由此我们可以利用 QR 分解求解 $\hat{\beta}$:

$$\begin{aligned}
\hat{\beta} &= (X^\top X)^{-1}X^\top Y \\
&= (R^\top Q^\top QR)^{-1}R^\top Q^\top Y \\
&= (R^\top R)^{-1}R^\top Q^\top Y \\
\Rightarrow \quad (R^\top R)\hat{\beta} &= R^\top Q^\top Y \\
\Rightarrow \quad R\hat{\beta} &= Q^\top Y \qquad\qquad (R \text{ 为 square } \Rightarrow (R^\top)^{-1} \text{ 存在})
\end{aligned}$$

可以使用 backward substitution 快速求解

2 FWL Theorem 的 Application: Partial Correlation

Logic

考虑 Pearson's correlation:

$$\hat{\rho}_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \in [-1, 1]$$

其与 simple regression $y \sim x$ 有紧密联系:

$$\hat{\beta}_{\text{simple regression}} = \frac{\hat{\text{Cov}}(x, y)}{\hat{\text{Var}}[x]} = \hat{\rho}_{xy} \frac{\hat{\sigma}_y}{\hat{\sigma}_x} \propto \hat{\rho}_{xy}$$

我们希望研究 multiple regression 是否也和 correlation 有类似的联系

2.1 Partial Correlation 和 Multiple Regression 的联系

Settings:

考虑以下 multiple regression:

$$y = (y_1, \ldots, y_n)^\top \qquad X = (\vec{1}, x, w)$$

其中 $w \in \mathbb{R}^{n \times k}, k \geq 1$

我们仍希望将 $\hat{\beta}_x$ (x 对应的系数) 与 x 和 y 的 correlation 联系起来

Partial correlation 的引出:

考虑以下两个 multiple regressions:

$$\begin{aligned} &\text{Regression 1: } y \sim 1 + w \quad \Rightarrow \quad \begin{cases} \text{residuals: } \hat{\varepsilon}_{y_i} \\ \text{RSS: } RSS_y = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \end{cases} \\ &\text{Regression 2: } x \sim 1 + w \quad \Rightarrow \quad \begin{cases} \text{residuals: } \hat{\varepsilon}_{x_i} \\ \text{RSS: } RSS_x = \sum_{i=1}^n (x_i - \hat{x}_i)^2 \end{cases} \end{aligned}$$

则可以定义 partial correlation $\hat{\rho}_{yx|w}$ 为:

$$\hat{\rho}_{yx|w} := \hat{\rho}_{\hat{\varepsilon}_y \hat{\varepsilon}_x} = \frac{\sum_{i=1}^n \hat{\varepsilon}_{x_i} \hat{\varepsilon}_{y_i}}{\sqrt{RSS_y} \cdot \sqrt{RSS_x}}$$

Remark: intuition

由于:

- $\hat{\varepsilon}_y$ 表示 y 中无法被 w 解释的部分
- $\hat{\varepsilon}_x$ 表示 x 中无法被 w 解释的部分

因此它们的 correlation 即表示剔除 w 的影响后, x 与 y 的相关性

Partial correlation 和 multiple regression 的联系:

考虑以下 simple regression:

$$\hat{\varepsilon}_y \sim 1 + \hat{\varepsilon}_x$$

则根据 simple regression 的性质, 有:

$$\hat{\beta}_{\hat{\varepsilon}_x \hat{\varepsilon}_y} = \hat{\rho}_{xy|w} \sqrt{\frac{RSS_y}{RSS_x}}$$

根据 FWL theorem, $\hat{\beta}_x = \hat{\beta}_{\hat{\varepsilon}_x \hat{\varepsilon}_y}$, 因此

$$\hat{\beta}_x = \hat{\rho}_{xy|w} \sqrt{\frac{RSS_y}{RSS_x}}$$

2.2 Theorem: Partial Correlation 的 Explicit Form

令 $X, Y, W \in \mathbb{R}$, 则 partial correlation 可被表示为

$$\hat{\rho}_{XY|W} = \frac{\hat{\rho}_{XY} - \hat{\rho}_{YW}\hat{\rho}_{XW}}{\sqrt{1-\hat{\rho}_{YW}^2}\sqrt{1-\hat{\rho}_{XW}^2}}$$

Remark: Simpson's paradox

即便 X 和 Y 为 negatively correlated, $\hat{\rho}_{XY|W}$ 仍可能为 positive

考虑 $\hat{\rho}_{YW}$ 为 significantly positive, $\hat{\rho}_{XW}$ 为 significantly negative 的情况, 则 $\hat{\rho}_{XY|W}$ 可能为 negative

Example: Model Grad School Admission Decisions at Berkeley

Data:

All students applying to Berkeley this year:

- $Y_i = \begin{cases} 1, & \text{if student is admitted} \\ 0, & \text{if student is not admitted} \end{cases}$
- $X_i = \begin{cases} 1, & \text{if student is female} \\ 0, & \text{otherwise} \end{cases}$
- $W_i = \text{admission rate from last year for departments where student applies}$

Models:

考虑以下两种 models:

- Model 1: $Y \sim 1 + X$
- Model 2: $Y \sim 1 + X + W$

Findings:

- Model 1 中 X 的 coefficient 为 negative (即 $\rho_{XY} < 0$), 表示女性劣势
- Model 2 中 X 的 coefficient 为 positive (即 $\rho_{XY|W} > 0$), 表示女性优势

造成差异的原因可用 Simpson's paradox 解释, 即女性倾向于申请 admission rate 较低的项目 ($\rho_{XW} < 0$), 而 admission rate 和 admission 呈正相关 ($\rho_{YW} > 0$), 因此即便 $\rho_{XY} < 0$, 仍然有 $\rho_{XY|W} > 0$